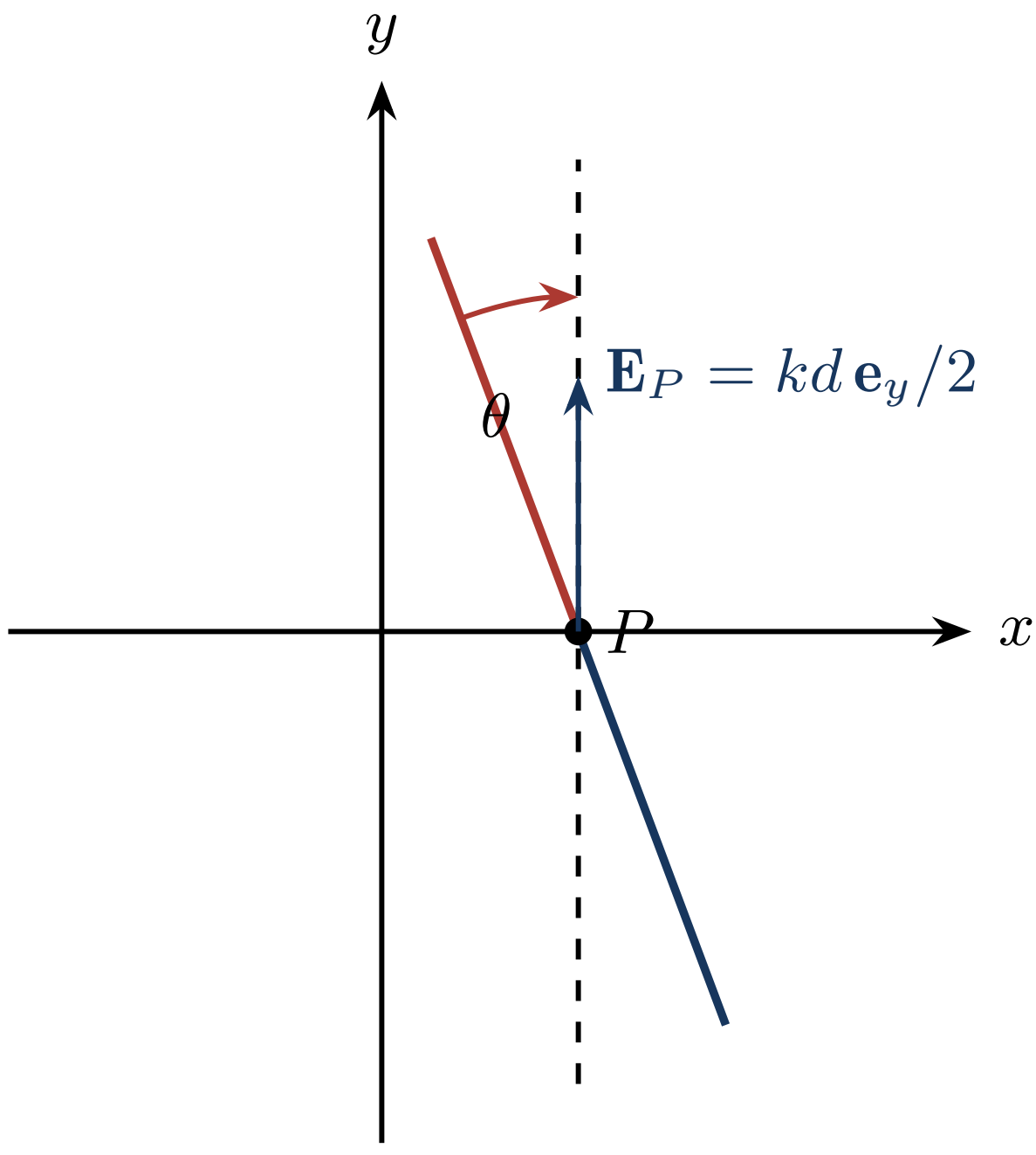

上端向左偏离竖直时，$\tau_P = -kQld\sin\theta/4$，力矩使杆回到竖直。